

NATIONALLY DETERMINED CONTRIBUTIONS

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HISTORY OF THE UNITED ARAB EMIRATES



SCAN FOR
SOURCES



THE UAE, A FEDERATION OF EMIRATES OR ARAB STATES, WAS FORMED IN 1971, IT IS A GLOBAL OIL LEADER, IT SERVICES HUB, AND TRADE CENTER, HOSTING A POPULATION OF ~11 M & OVER 200 NATIONALITIES.

GIVEN THIS DIVERSITY OF ECONOMIC ACTIVITY, RESIDENT & INTERNATIONAL STAKEHOLDERS, BUILDING ROBUST SYSTEMS THAT ARE **FUTURE-PROOF** AND **CLIMATE RESILIENT** IS OF THE **UTMOST PRIORITY**. THIS PRESENTATION IS FOCUSED ON INFORMING & ENCOURAGING INVESTORS TO BANK ON THE UAE'S TRANSITION TO A LOW CARBON FUTURE, & THE MANIFOLD GAINS IT WILL PRESENT.

ECONOMIC DIVERSIFICATION



75%

OF THE ECONOMY
COMES FROM NON OIL
SECTORS



Data source: Global Carbon Budget (2025); Population based on various sources (2024)
OurWorldinData.org/co2-and-greenhouse-gas-emissions | CC BY

1. Fossil CO₂ emissions This refers to the carbon dioxide released when burning fossil fuels or from certain industrial activities. Burning fossil fuels – coal, oil, and gas – produces CO₂ during transport (cars, trucks, planes), electricity generation, heating, and energy use in industry. This also includes flaring, which is the burning of extra gas during oil and gas extraction. Some industrial processes also release CO₂. This happens especially in cement and steel production, where chemical reactions (unrelated to burning fuel) produce carbon dioxide. These figures don't include CO₂ emissions from changes in land use, like deforestation or reforestation.



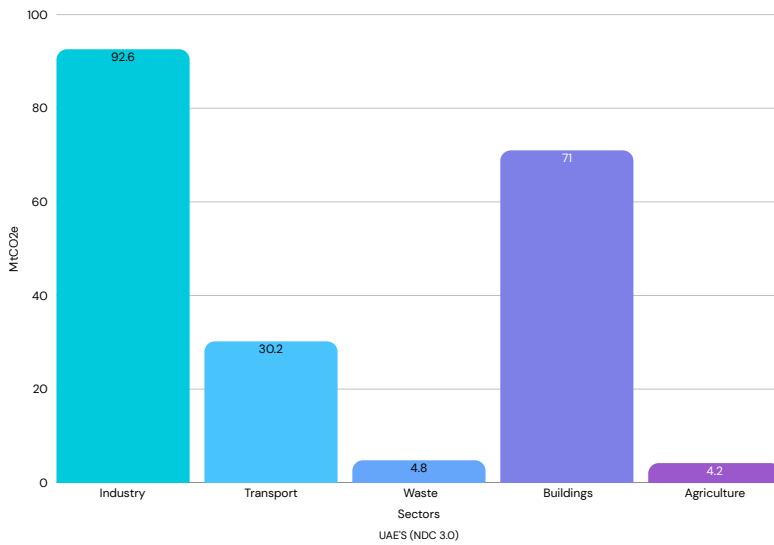
HISTORICALLY, THE UAE'S EXPORTS HAVE BEEN DOMINATED BY FOSSIL FUEL PRODUCTS INCLUDING CRUDE AND REFINED PETROLEUM (BOTH LIQUID & GAS), AND GOLD (A CONFLICT METAL). THESE INDUSTRIES HAVE HIGH SOCIAL & ENVIRONMENTAL COSTS & IMPACTS - HENCE, SINCE 2017, THE UAE HAS BEEN FOCUSING ON ECONOMIC DIVERSIFICATION DRIVEN BY ITS VISION FOR A NATIONAL CLIMATE PLAN (2017 TO 2050) (AS OUTLINED IN UAE'S NDC FRAMEWORK).

THE NON-OIL SECTOR TODAY ACCOUNTS FOR APPROXIMATELY 75% OF GDP, UP FROM ONLY 30-40% IN THE 1970S. THIS INCLUDES HOSPITALITY, REAL ESTATE, INDUSTRIAL, CONSTRUCTION, AND TRANSPORTATION.



SECTORAL PATHWAY OF UAE

Sectoral Emission
2019



UAE SECTORAL EMISSIONS FRAMEWORK AND REDUCTION TARGETS :

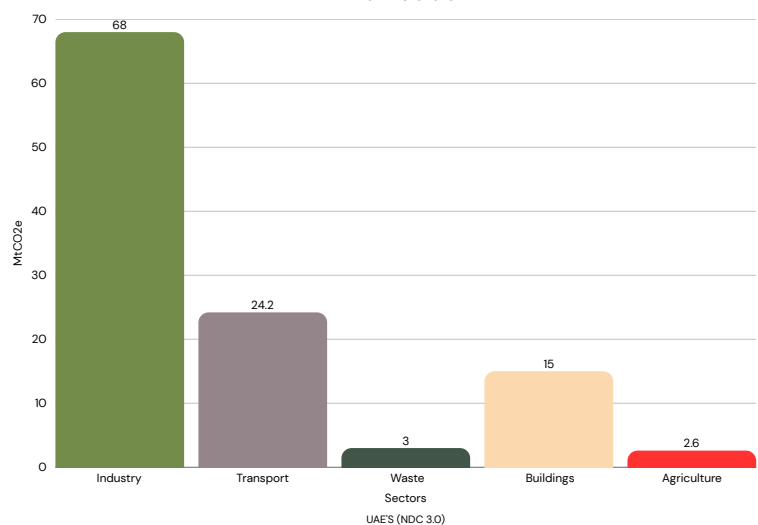
THE UAE'S DIVISIONAL EMISSIONS ACCOUNTING ABIDES BY IPCC GUIDELINE AND UPHOLDS THE NET ZERO BY 2050 STRATEGIC INITIATIVE, ENSURING TRANSPARENCY AND CLARITY AND RESPONSIBILITY.

EMISSIONS FROM WATER GENERATION AND ENERGY (FOSSIL FUELS) ARE DOCUMENTED WITHIN THAT SECTOR BUT ALSO SHARED AMONG END-USE SECTORS: TRANSPORT, BUILDING, WASTE, INDUSTRY AND AGRICULTURE TO INDICATE EMISSION AND ENCOURAGE ENERGY PRODUCTIVITY AND SHIFTING IN ENERGY SOURCE.

THIS DUAL ACCOUNTING METHOD GUARENTEES EACH SECTOR CARRIES ACCOUNTIBILITY FOR ITS ENTIRE GHG IMPACT AND ORIGRESS CAN BE CLEARLY MEASURED

IFUNDAMENTALLY THE UAE'S EMISSIONS FRAMEWORKS STIMULATES INTERDIVISIONAL ACCOUNTABILITY , MONITORS BOTH DIRECT AND INDIRECT EMISSIONS AND INCORORATES NATUTRAL AND TECHNOLOGICAL CARBON SINKS, GUAREANTEENG A RATIONAL APPROACH TOWARDS ITS 47% REDUCTION BY 2035 AND NETZERO BY 2050

Sectoral Reduction Targets by 2035
From 2019 level



ECONOMIC RATIONALE



COSTS OF INACTION



PHYSICAL RISKS



TRANSITION RISKS



OPPORTUNITIES

BUILDINGS & INDUSTRIES



- Higher bills for water & power consumption due to increased need for desalination, ambient & industrial cooling
- Higher prices for traditional building & raw materials



- Threats to coastal infrastructure due to sea level rise and increased frequency of storms; unprecedented inland floods & associated building damage
- Obsolete designs may pose safety risks to residents/workers during climate disasters



- Evolving green standards might lead to costly building retrofitting & fines for high-emission construction materials & inefficient manufacturing processes
- Stringent standards can cause delays or cancellations of non-compliant projects



- Raising finance for green building materials, modular or prefabricated structures, investing in green hydrogen and clean energy like wind & solar.
- Building advanced wastewater recycling, pollutant capture systems; adopting traditional cooling techniques in building design



- Global sanctions imposed for poor rates of recycling and excess emissions from high rates of landfill disposal
- Poor segregation practices attract criticism in the press & social media



- Heavy metal contamination from e-waste affects public safety & the environment, even if the waste is exported to other countries for processing
- Nuclear waste storage facilities are vulnerable to climate disasters like cyclones/floods

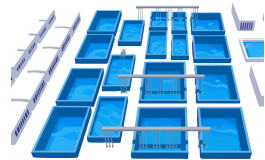


- Developing countries can stop or raise the costs of importing waste from the UAE which damages their local environments, leading to a pileup of waste within the country
- Low availability of land for landfills can lead to buildup of undisposed waste



- Building energy-efficient waste processing & waste-to-energy units, achieving higher rates of critical metal extraction from e-waste & generate employment from segregation work
- Use recycled plastics as construction material for infrastructure

WASTE MANAGEMENT



- High costs of water purification & recycling for cooling systems
- High emissions from air conditioning



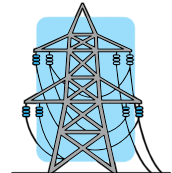
- Acute water scarcity can bring the economy to a standstill
- Extreme heat events strain the electricity grid, decrease efficiency



- Drop in demand for thermal-based power causes asset obsolescence, economic losses



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POWER

WATER RESOURCES

AGRICULTURE



- Increasing costs incurred by farmers to create optimal growing conditions & gain optimal productivity in the face of rising temperatures, water scarcity, & extreme weather events.
- Increasing risks of crop damage, pest infestations, pollinator decline, & desertification of arable lands due to climate change causing erratic wind & rainfall patterns.
- Global shifts towards lower pesticide & chemical usage & the practice of polycultures versus monocultures can damage the competitiveness & green credentials of produce.
- Investing in crop diversification, pollinator revival, organic, sustainable, & traditional smaller-scale farming practices, efficient irrigation methods using recycled water, & offsetting emissions via agroforestry.



TRANSPORTATION



- Increasing costs of maintaining fossil fuel fleets in the face of rapidly evolving electric- & hydrogen-fuelled fleets.



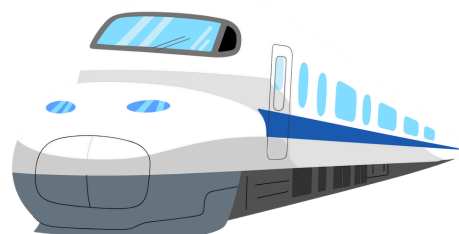
- Increased floods & extreme heat pose risk to roads, tyres, & traditional vehicle parts which can overheat & face permanent damage in commercial, public, & private vehicles.
- Changing winds & upper atmospheric activity can carry aerosols from different parts of the world to obscure visibility for aeroplanes, cause turbulence, & stress internal systems.



- Global shifts towards biofuels & other sustainable fuels such as recycled cooking oil - will make engines designed for fossil fuels dysfunctional, necessitating complicated upgrades.



- Invest in research & development towards building charging stations for electric fleets, battery recycling units, improving the efficiency of solar panels, and encouraging household & commercial energy harnessing via wind & solar.
- Building strong public transportation networks with high connectivity & climate resilient infrastructure.



BUSINESS RATIONALE

WITH ITS THRIVING TECHNOLOGICAL & FINANCIAL, AS WELL AS MANUFACTURING, TRANSPORTATION, CONSTRUCTION, & MANY OTHER SECTORS, THE UAE HAS THE OPPORTUNITY TO SECURE A FUTURE FOR ITS WORKING POPULATION THAT IS JUST, AND CUSHIONED AGAINST THE ADVERSE EFFECTS OF CLIMATE CHANGE THROUGH FORESIGHT & RESPONSIBLE PRE-EMPTION. THIS WAY, THE FEDERATION WILL STILL CONTINUE TO ATTRACT TOP TALENT FROM AROUND THE WORLD & REMAIN A COVETED DESTINATION FOR GLOBAL COMMERCE, INFORMATION EXCHANGE, PIONEERING RESEARCH, EDUCATION, & EMPLOYMENT GENERATION (REFERENCE: NDC DOCUMENT).



POWER & WATER SECTOR

MASDAR & ACWA POWER ARE RENEWABLE ENERGY LEADERS & DESALINATION EXPERTS WHO ARE IN THE FOREFRONT OF UAE'S TRANSITION TO A GREEN ECONOMY IN ALIGNMENT WITH ITS PARIS 2015 COMMITMENTS AND 2050 NET ZERO AMBITION (SOURCES IN QR).

THE RENEWABLES SECTOR WILL ALSO DRIVE THE ADOPTION OF CLEANER ENERGY SOURCES AMONG THE UAE'S OTHER ENERGY-INTENSIVE HEAVY INDUSTRIES - LIKE ALUMINIUM, SHIP-BUILDING, LOGISTICS, & TELECOM. IT WILL DRIVE STIFF COMPETITION TO THE PETROCHEMICALS PROCESSING & THERMAL ENERGY SECTORS, OUTCOMPETING THEM & GREENING THE GRID IN THE LONG RUN, WITH THE RIGHT POLICY INCENTIVES AND FINANCING, PROGRESS CAN ACCELERATE.

TRANSPORT

TESLA, BYD, KIA, GENERAL MOTORS, & MENA MOBILITY ARE AMONG THE MAJOR EV PLAYERS WHO ARE DOMINATING THE UAE MARKET IN TERMS OF DEVELOPING CHARGING INFRASTRUCTURE & MARKETING ELECTRIC CARS. APP-BASED CAR RENTAL MODELS ARE ALSO BEING PIONEERED BY SEVERAL STARTUPS IN THE EV SPACE, LIKE ESYASOFT & CARASTI.

WASTE MANAGEMENT

AVERDA, BEE'AH, TADWEER, & DULSCO ARE SOME OF THE WASTE LEADERS WHO PROVIDE SEGREGATION, WASTE-TO-ENERGY, & RECYCLING FACILITIES. THESE COMPANIES WORK CLOSELY WITH INDUSTRIES, & WILL BENEFIT GREATLY FROM BOTH DIRECT INVESTMENTS AS WELL AS ESG FUNDING FROM LARGE COMPANIES LIKE DP WORLD, ABU DHABI SHIP BUILDING, DUBAI CABLE COMPANY, ETISALAT, MUBADALA INVESTMENT, & OTHERS WHO LEAD THE INDUSTRIAL SPACE.



AGRICULTURE

DUE TO THE SCARCITY OF WATER IN THE UAE, PROMINENT AGRICULTURE-FOCUSED COMPANIES LIKE AL DAHRA, ELITE AGRO PROJECTS LLC, PURE HARVEST SMART FARMS, & EMIRATES HYDROPONICS FARMS ARE PIONEERING THE USE OF WATER-EFFICIENT TECHNIQUES AND MINIMISING CHEMICAL ADDITIVES IN FARM PRODUCE. BEING AN ARID REGION, THE UAE IS PRIMARILY FOCUSED ON ACHIEVING FOOD SECURITY FOR ITS GROWING POPULATION AND INFLUX OF IMMIGRANTS - WHILE ALSO STRIVING TO GROW ITS AGRICULTURAL EXPORTS MARKET.

INVESTMENTS IN THIS SECTOR CAN PROVIDE OPPORTUNITIES TO INNOVATE WITH LOW-RESOURCE & RESILIENT VARIETIES OF CROPS WHICH CAN WITHSTAND FUTURE CLIMATE VARIABILITIES, HIGHER TEMPERATURES, & CONTINUE TO CONTRIBUTE TO THE GLOBAL SUPPLY OF VEGETABLE & SEED OILS, DATES, SPICES, COFFEE, MEAT & DAIRY, WINE, & MORE.

SOCIETAL RATIONALE



IT WILL CREATE JOBS AND NEW SECTORS:
SUCH AS SUSTAINABLE BUSINESS REFORM-POWER SAVING PRODUCTION, GREEN FUEL LOGISTICS, PRODUCTION OF ELEMENTS FOR SUSTAINABLE POWER WHICH HAS POTENTIAL TO FORM FRESH TRAINED JOBS ROLES PRIMARILY YOUNGER GENERATIONS

ACCESS AND EQUITY: HAVING DEPENDABLE, INEXPENSIVE AND CLEAN UTILITIES REDUCE INEQUALITIES. WHICH GUARENTEES ALL INHABITANTS HAVE ACCES TO ESSENTIAL ENERGY, WATER WHICH UPHOLD SOCIETAL WELLBEING.

STENTHEN SOCIETY: GREEN ENERGY AND EFFICIENT WATER SUPPLY REINFORCE COMMUNITIES BY REDUCING FRAGILITY TO CLIMATE STRESS(WATTERSCARCITY/HEAT) WHICH SECURING STABLE UTILITY REACH DURING ENVIRONMENTAL CRIES AND RAISING PUBLIC HEALTH

HUMAN RIGHTS ALIGNMENT:
FUNDNG IN SUTAINABLE AND AND SECURE UTILITIES UPHOLD THE FUNDAMENTAL RIGHT TO CLEAN WATER AND ENERGY, CORRESPONDING WITH THE UAE'S COMMITMENT TO SOCIAL WELFARE AND ITS SUSTAINABLE ECONOMIC PLAN

ECONOMIC EQUITY AND INCLUSION:
BROADENING THE INDUSTRIAL BASE IS ABLE TO REDUCE OVER RELIANCE ON FOSSIL FUEL WORK- FACILITATING TO DEMOCRATIZE ECONOMIC OPPERTUNITIES THROUGHOUT VARIOUS SECTION OF THE SOCIETY.



COMMUNITY STABLE AND RESILIENCE:
INDUSTRY DRIVEN BY RENEWABLES AND EFFICIENT RESOURCES MANAGEMENT REDUCES ECOLOGICAL CONTAMINATION AND OVEREXPLOITATION (WATER ,ENERGY) WHICH AIDS EMPLOYEE WELL-BEING AND NEIGHBOURING POPULATIONS, FOSTERING A HEALTHIER HABITAT AND BETTER STANDARD OF LIFE.

SOCIL-GOVERNNCE BENEFITS:

AS INDUSTRIES IMPLEMENT ECOFRIENDLY METHODS,LEGAL CONFORMITY ,SAFEWORKING CONDITION AND ECOLOGICAL BENCHMARKS BECOME MORE ROBUST , ALIGNING WITH MORE EXTENSIVE HUMAN RIGHTS AND SOCIETAL RULES . THIS UPHOLDS PULIC PERMISSION TO FUNCTION LONG TERM SOCIETAL AGREEMENT.

CLEAN ENERGY SUPPLY CHAINS AND GREENMANUFACTURING CREATE NEW SKILLD JOBS

FOSTERING INCLUSIVE ECONOMIC PARTICIPATION,REDUCES OIL DEPENDENCY. GREEN PRODUCTION IMPROVES LABOURS'S HEALTH AND WELL-BEING OF COMMUNITY. STRENGTHENS GOVERNANCE,LABOUR STANDARDS AND SOCIAL LICENCE TO OPERATE

ELECTRIC VEHICLE AND PUBLIC TRANSPORT SYSTEMS CREATE VARIED JOB OPENINGS .BROADENS COST EFFECTIVE ACCESS FOR ALL SOCIETAL DEMOGRAPHIC.REDUCE AIR POLLUTION. GUARENTEE FAIR RESILIENT TRANSPORT ACESS OVER THE COURSE OF ENVIRONMENTAL AND ENERGY EMERGENCIES.

RECYCLING WASTE TO ENERGY SECTOR GENERATE GREEN JOBS. LOWERS AIR POLLUTION AND HEALTH RISKS IN VULNERABLE POPULATIONS, FOSTERS MATERIAL RECLAMATION BOSOSTING DURABILITY. VALIDATES UNORGANIZED WORKERS,GUARANTEETING EQUITABLE INCOMES/SUPPORT.



CLIMATE FINANCE AIMS TO DELIVER NOT ONLY ENVIRONMENTAL BENEFITS BUT ALSO ESSENTIAL SOCIAL BENEFITS SUCH AS EMPLOYEMENT,WELFARE,EQUITY AND RESILIENCE, THEREBY CREATING SHARED VALUE. THESE INVESTMENTS SRTHEN THE UAE'S LONG TERM SOCIAL COMPACT, WHICH INCLUDES ENSURING ACCESS TO BASIC SERVICES SUCH AS ENERGY WATER, HOUSING AND FOOD TO MAINTAIN SOCIAL INCLUSION AND STABILITY EVEN DURING CLIMATE CHANGE. ADDITIONALLY, THEY ENHANCE COMMUNITY AND SOCIETAL RESILIENCE,MAKING PEOPLE LESS VULNERABLE TO THREATS FROM CLIMATE SHOCKS, RESOURCES STRESS AND POLLUTION. THUS FOR INVESTORS, THIS FUNDING IS NOT LIMITED TO CARBON REDUCTION ALONE, BUT ALSO PROVIDES A GREATER RETURN ON SOCIAL CAPITAL BY SUPPORTING SUSTAINABLE DEVELOPMENT GOALS(SDGS) AND LONG TERM HUMAN WELL BEING.